Socartes RAG Capability Test on *The Haunted Pajamas*

Abstract

This report evaluates whether Socartes can answer obscure fiction questions from a supplied local corpus while refusing unsupported questions. The main Socartes condition uses the full Project Gutenberg text of *The Haunted Pajamas* automatically split into 1047 chunks.

The repaired full-novel retriever uses hybrid recall with Reciprocal Rank Fusion (BM25 top 200 plus dense-style semantic recall top 200), a 300-candidate rerank pool, cross-encoder-style reranking, `top_k=5`, `adjacent_hops=1`, title-term removal, and a direct-support refusal gate. It answers the original 10 of 10 source-supported questions and 30 of 30 extension questions, for 40 of 40 answerable questions overall, while correctly refusing 2 of 2 unsupported controls.

Socartes RAG: Full-Novel Capability Test

42-question diagnostic; full-novel RAG uses hybrid RRF reranking and a direct-support gate.

Condition	Answerable	Q11-Q12	Role
GPT-5.5 closed-book probe No search, no database	2/10	1/2	reference
Baseline: legacy top-1 retrieval Single highest lexical-overlap chunk	22/40	1/2	baseline
Socartes full-novel automatic index 1047 deterministic automatic chunks	40/40	2/2	main test

Hybrid RRF plus reranking reaches 40/40 answerable questions while the support gate keeps refusals at 2/2.

Research Question

Can the Socartes RAG retriever, using the same full-novel corpus but a repaired hybrid retrieval pipeline, retrieve the right evidence from a full novel and enforce refusal when the local corpus does not support the question?

Corpus

The corpus is Project Gutenberg EBook #33780, *The Haunted Pajamas* by Francis Perry Elliott:

- Book page: https://www.gutenberg.org/ebooks/33780
- Plain text source: https://www.gutenberg.org/ebooks/33780.txt.utf-8

- Local fixed copy: `experiments/data/haunted_pajamas_33780.txt`

The script removes the Project Gutenberg start/end boilerplate, then chunks the remaining body by paragraph accumulation with `chunk_target_words=100`. With the committed text this yields:

Item	Value
Body word count	80,408
Automatic chunk count	1,047
Retrieval system	`StoryRagIndex` with hybrid RRF recall, reranking, and adjacent evidence aggregation
Tie break	same-score chunks choose the earliest chunk position

Conditions

Condition	Role
GPT-5.5 closed-book probe	Existing reference data; no search and no database access.
Baseline: legacy top-1 lexical retrieval	Previous Socartes retrieval setting before the top-k adjacent evidence repair.
Socartes full-novel automatic index	Main experiment over the full automatic index.

Questions

The diagnostic now has 42 questions. Q1-Q10 are the original answerable regression set, Q13-Q42 are answerable full-novel detail questions, and Q11-Q12 are unsupported controls.

ID	Expected behavior
Q1	Retrieve the package sender/name-address evidence.
Q2	Retrieve the Carlton/Mastermann identity evidence.
Q3	Retrieve the red silk muffler evidence.
Q4	Retrieve the unpaid-debt evidence.
Q5	Retrieve the Hickey's Pride cigar evidence.
Q6	Retrieve the two-for-five evidence.
Q7	Retrieve the suit-of-pajamas evidence.
Q8	Retrieve the Old Memphis Tuffles evidence.
Q9	Retrieve the little-spider evidence.
Q10	Retrieve the tarantula evidence.
Q11	Refuse the Sherlock Holmes question.
Q12	Refuse the spaceship-captain question.

ID	Expected behavior
Q13	Retrieve the Boston-arrival evidence.
Q14	Retrieve the little-red-snakes cord evidence.
Q15	Retrieve the Foxy impostor evidence.
Q16	Retrieve the Doozenberry distinguished-scientist evidence.
Q17	Retrieve the Si-Ling-Chi lost-silk evidence.
Q18	Retrieve the red-whale car comparison.
Q19	Retrieve the five-hundred-dollar reward evidence.
Q20	Retrieve the Captain Clutchem evidence.
Q21	Retrieve the chauffeur-under-the-pergola evidence.
Q22	Retrieve the ruby/library evidence.
Q23	Retrieve the good-excalibar evidence.
Q24	Retrieve the Miss-Billings correction evidence.
Q25	Retrieve the judge's my-son claim evidence.
Q26	Retrieve the golden-humming-birds dream evidence.
Q27	Retrieve Jenkins' scrimmage wording.
Q28	Retrieve the phusiotus bug guess.
Q29	Retrieve the phanaeus-carnifex bug identification.
Q30	Retrieve the Sleepy Hollow and Pocantico Hills drive evidence.
Q31	Retrieve the fragrant-pine-cones hearth evidence.
Q32	Retrieve the Manchurian railway evidence.
Q33	Retrieve the burglar's "mine now to keep forever" ruby evidence.
Q34	Retrieve the petticoat-unknown-in-China evidence.
Q35	Retrieve Wilkes' "jimmies" description of Billings.
Q36	Retrieve Lightnut's plan to send the pajamas to Billings.
Q37	Retrieve the four-curs dog-fight detail.
Q38	Retrieve the "disgrace to an honored name" hallway evidence.
Q39	Retrieve Frances' lovely-crimson face and neck evidence.
Q40	Retrieve the young fellow's "a bit sulky" description.
Q41	Retrieve the professor's proposed call on Billings.
Q42	Retrieve the narrator's near loss of consciousness.

Main Results

Condition	Original Q1-Q10	New Q13-Q42	Overall answerable	Correct refusal Q11-Q12	Source form
GPT-5.5 closed-book probe	2/10	not run	2/10	1/2	none
Socartes full-novel automatic index	10/10	30/30	40/40	2/2	chunk ID

The main RAG result after adding Q33-Q42 is 40/40 answerable retrieval and 2/2 correct refusal on the full-novel automatic index.

Repair Ablation

Configuration	Original Q1-Q10	New Q13-Q42	Overall answerable	Correct refusal Q11-Q12
Baseline: legacy top-1 lexical retrieval	7/10	15/30	22/40	1/2
Hybrid RRF + rerank without support gate	10/10	30/30	40/40	1/2
Full setting: hybrid RRF + rerank plus support gate	10/10	30/30	40/40	2/2

The ablation separates the two failure modes. The legacy top-1 baseline misses Q2, Q5, and Q10 because the answer phrase is in a lower-ranked or adjacent chunk. Hybrid RRF recall and reranking recover those answer passages and the harder Q13-Q42 extension set. The direct-support gate then fixes Q12 by refusing to answer the spaceship-captain question from an unrelated Captain Clutchem police context. The added Q13-Q42 extension shows why simple top-k expansion is not enough: several correct passages are outside the lexical top-100 window or require relation matching rather than direct token overlap.

Baseline Error Analysis

Within the original Q1-Q10 set, the failed answerable questions in the legacy top-1 baseline are Q2, Q5, and Q10. They are not coverage failures; the evidence exists in the automatic index. They are retrieval-window failures caused by selecting only one lexical-overlap chunk. Across all 40 answerable questions, the same legacy baseline reaches 22/40, leaving 18 retrieval misses that the hybrid pipeline recovers.

ID	Chosen chunk	First evidence rank	Interpretation
Q2	`full-novel-auto-0004`	23	The Carlton evidence is present, but several earlier chunks tie or beat it on generic overlap with `Jenkins` and `Mastermann`.
Q5	`full-novel-auto-0019`	4	The Hickey's Pride evidence is near the top but misses top-3 under pure word-overlap ranking.

ID	Chosen chunk	First evidence rank	Interpretation
Q10	`full-novel-auto-0041`	30	The tarantula evidence is nearby but receives only one overlapping query term, while earlier pajama-leg chunks tie at score 2.

Q12 fails in the no-support-gate ablation for a different reason. The novel contains a real `Captain` name, Captain Clutchem, in an unrelated police context. The question asks for a spaceship captain, but token overlap sees `captain` and `name`, reaches the refusal threshold, and returns a false hit. The repaired support gate requires the target terms from `name of the spaceship captain` to appear in the retrieved evidence, so it refuses because `spaceship` is absent.

Extended-Question Error Analysis

Before the hybrid repair, Q16, Q17, Q19, Q21, and Q22 failed even though each expected answer existed in the automatic full-novel index. Their first matching evidence was outside the final evidence window: Q16 and Q17 first appeared at rank 7, Q21 at rank 16, Q22 at rank 10, and Q19 at rank 62. The later Q23-Q32 extension exposed the next limit of pure lexical recall. Q23 first appeared at lexical rank 287, Q27 at rank 542, and Q29 at rank 67. Q24 appeared at rank 29 but required the reranker to prefer the exact "Miss Billings" correction passage over generic Frances/Lightnut context. Hybrid RRF recall brings semantically related evidence into the candidate pool, and cross-encoder-style reranking selects the passage that directly answers the question. The added Q33-Q42 diagnostic stays correct under the same pipeline.

Interpretation

This experiment shows that Socartes can provide verifiable chunk IDs and correct refusal behavior on a full corpus when retrieval combines lexical recall, semantic recall, reranking, adjacent evidence aggregation, and a direct-support gate.

The main claim should stay scoped to this implementation and corpus: Socartes can ground 40 of 40 answerable diagnostic questions in a local full-novel index and refuse the two unsupported controls. Broader robustness still requires follow-up work such as larger multi-corpus tests, production embedding backends, learned cross-encoder reranking, and stronger semantic support checking.

Reproducibility

Run the deterministic experiment:

```
python experiments/full_novel_eval.py
```

The script writes:

```
experiments/results/full_novel_eval.json
```

Focused RAG unit tests still pass:

```
pytest tests/test_story_rag.py -q
```

Relevant files:

- `experiments/full_novel_eval.py`
- `experiments/data/haunted_pajamas_33780.txt`
- `experiments/results/full_novel_eval.json`
- `backend/socartes_backend/story_rag.py`
- `tests/test_story_rag.py`